Endogenous Education Choice in Segmented Search Markets

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Abstract

How do workers decide whether to acquire education? The standard answer weighs wages against costs, but this misses a deeper aspect of the choice: education is not just about whether the skilled market pays more, it is about whether that market will actually have jobs available after training that were worth the trouble. This paper develops a general equilibrium search model in which the fraction of workers who choose to train is determined endogenously alongside wages, unemployment, and market tightness in both markets. Workers differ in ability, which governs the cost of training; the skilled market additionally features on-the-job search and a job ladder, giving trained workers richer career dynamics than their unskilled counterparts. The model is estimated on U.S. data across multiple aggregate states of the economy using Simulated Method of Moments. This allows me to trace how the economy — and the education margin — adjusts over time after an aggregate shock. I follow this with two applications that demonstrate the model's usefulness: the first studies the effect of more generous outside options (UBI or unemployment insurance) on unemployment duration and match quality; the second compares two education subsidy designs: flow transfers during training versus upfront cost reductions.

1 Related Literature

This paper connects five bodies of work.

Education choice in search models

The most direct predecessors are the papers that embed education decisions in search-and-matching frameworks. Charlot and Decreuse (2005) build a two-sector model in which workers self-select into educated and uneducated pools; both markets clear simultaneously, and a composition externality arises because the marginal entrant degrades the mean ability of the pool she joins, depressing vacancy creation for all co-searchers. Charlot et al. (2005) study how education raises both adaptability and productivity in a single frictional market. Charlot and Decreuse (2010) extend the two-sector framework to two-dimensional heterogeneity—ability and schooling cost—and show that the rich over-invest in education while the poor under-invest. Charlot and Malherbet (2013) add employment protection and show that a direct training subsidy can correct the hold-up distortion that search frictions create for education investments. Wasmer (2006) studies the choice between general and specific skills in a single Mortensen–Pissarides market with firing costs.

These papers share a common limitation: the skilled market has no internal career dynamics. Once a worker enters the educated pool, wages are set by Nash bargaining and the match persists until exogenous destruction. There is no on-the-job search, no job ladder, and no mechanism through which trained workers climb toward better matches over time. The present paper adds exactly this structure. The skilled market features on-the-job search and a job ladder à la Cahuc et al. (2006), so that the value of training depends not only on the entry wage but on the entire trajectory of wage gains from climbing the ladder. This interaction between the education margin and job-ladder dynamics is absent from the existing education-in-search literature.

Search and matching foundations

The model builds on the search-and-matching tradition of Diamond (1982) and Pissarides (2000), with endogenous job destruction from Mortensen and Pissarides (1994) governing match dissolution in the unskilled sector. The job-ladder dynamics in the skilled sector rest on the framework of Postel-Vinay and Robin (2002), who show how on-the-job search and between-employer competition generate equilibrium wage dispersion. Cahuc et al. (2006) embed Nash bargaining into this environment—giving workers positive bargaining power while retaining the job-ladder structure—and provide the wage-determination protocol that

I adopt.

Robin (2011) extends the job-ladder model to include aggregate shocks and shows that endogenous job destruction coupled with worker heterogeneity amplifies productivity shocks, a mechanism that also operates in my model and generates countercyclical training incentives. Lise et al. (2016) and Lise and Robin (2017) introduce continuous worker and firm heterogeneity and study how sorting between the two sides evolves over the business cycle, but they take the type distribution in each market as exogenous. My paper endogenises it: the distribution of ability types among skilled workers is an equilibrium object, determined by the training cutoff.

Acemoglu (2001) studies a related mechanism in a different setting: when good and bad jobs coexist in a single search market, higher unemployment benefits shift the composition of employment toward good jobs by raising workers' reservation values. My model extends this logic to a two-market environment where the composition shift operates through the training margin rather than through firms' job-creation decisions within a single market.

Human capital in search models

A separate literature studies training and skill accumulation within a given search market, without the market-selection margin. Acemoglu and Shimer (1999) establish that search frictions create a hold-up problem for pre-market human capital investments: because wages are determined ex post, workers underinvest in education relative to the social optimum. This underinvestment result—confirmed in a richer environment by Charlot and Malherbet (2013)—motivates the policy exercises in Section **??**, where subsidies to training can correct the frictional distortion. Bagger et al. (2014) allow human capital to accumulate along the job ladder, and Audoly et al. (2024) build on this to quantify the contributions of job-ladder losses, skill depreciation, and firm-specific capital to the earnings losses of displaced workers. In these papers, training unfolds within a single frictional environment: it raises the worker's productivity or bargaining position but does not change which market the worker participates in.

This paper takes a different view. Training is a one-time, costly decision that grants permanent access to a qualitatively different labour market. The worker compares two entire continuation value functions, each governed by different search frictions, different turnover dynamics, and different equilibrium objects. This makes the training decision fundamentally more sensitive to general-equilibrium conditions than the within-match view implies. The model abstracts from within-market skill accumulation by treating ability x as a fixed

type. A natural extension would allow a skill multiplier to evolve with labour-market experience; Cohen et al. (2025) document that unemployment spells erode measurable skills independently of ability, providing empirical motivation for such an extension.

Education policy in general equilibrium

The competitive general-equilibrium literature on education policy—notably Heckman et al. (1998) and Abbott et al. (2019)—evaluates subsidies in OLG frameworks with competitive factor markets. In those models, the response to a subsidy operates through the skill premium: more educated workers raise relative supply and compress wages. Bárány (2016) adds endogenous directed technical change to this logic, showing that the education margin interacts with the direction of innovation. In all of these papers the relevant general-equilibrium channel is the relative price of skill. In my model, the channel is different. More trainees enter skilled unemployment, which affects the job-finding rate of all skilled unemployed workers by changing market tightness. This tightness feedback can amplify or dampen the direct effect of a subsidy in ways that competitive models cannot capture.

The policy exercise distinguishes two subsidy instruments: a flow transfer during the training spell and a one-time reduction in training costs. This distinction maps onto a design dimension studied empirically by Lindner and Reizer (2020), who show that the timing of benefit payments—flow versus frontloaded—generates different search responses holding total transfers constant. The exercise is also connected to the active labour market programme evaluation literature: Card et al. (2018) synthesise over 200 programme evaluations and find that training programmes generate larger positive effects in the medium run than in the short run, a pattern consistent with the job-ladder dynamics in the skilled sector, where newly trained workers enter at the bottom of the quality distribution and climb over time.

The empirical finding of Nekoei and Weber (2017) that more generous unemployment benefits improve reemployment match quality—not just search duration—provides the reduced-form motivation for the role of non-employment payoffs in the model: a higher b_U raises reservation match quality in the unskilled market, but it also compresses the net gain from training and shifts the training threshold, generating cross-market effects that a single-market framework cannot capture.

Countercyclical enrolment

A parallel empirical literature documents that college enrolment is countercyclical. Dellas and Sakellaris (2003) establish this pattern using four decades of US data. Clark (2011) con-

firms it for England using regional variation in youth unemployment. Barr and Turner (2015) show that the Great Recession triggered enrolment increases concentrated at non-selective institutions, and that unemployment-insurance benefit duration played a significant role in the timing of these decisions. Blom et al. (2021) find that community college enrolment responds to local labour-market conditions with an elasticity of approximately 0.1; Ulvestad and Skjelbred (2023) extend the evidence to graduate-level enrolment in Norway, confirming that the pattern is not specific to U.S. undergraduates. The model generates countercyclical enrolment endogenously: a recession lowers the contact rate in the unskilled market, compressing the continuation value of remaining unskilled and lowering the training threshold $\bar{x}(z)$, so that a larger mass of workers finds it optimal to train. This is the structural mechanism behind the aggregate enrolment patterns that the empirical literature documents.

Bičáková et al. (2021) and Bičáková et al. (2023) go further and show that graduates who enrol during recessions earn higher wages later, an effect that cannot be attributed to ability selection at entry. Their finding of a conditional wage premium operating through effort and preference formation during college lies outside the scope of the current model, which abstracts from the training spell itself.

2 Model

I develop a general-equilibrium search model in which workers endogenously choose whether to invest in training and thereby gain access to a higher-skill labour market. Training decisions, wages, unemployment, and market tightness are determined jointly in both markets, so the education margin responds not only to the skilled wage premium but also to the availability of skilled jobs after training. The education-choice structure draws on Charlot and Decreuse (2005, 2010), while the skilled-market dynamics build on the job-ladder framework of Postel-Vinay and Robin (2002) with the Nash-bargaining wage protocol of Cahuc et al. (2006).

2.1 Environment

Time is continuous and all agents discount the future at rate $r > 0$. A unit mass of infinitely lived workers, each characterised by an innate type $x \in [0, 1]$ drawn from a Beta(a_ℓ, b_ℓ) distribution with CDF L and density ℓ, populates the economy. Workers exit at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers drawn from the same distribution.

Workers participate in one of two labour markets—unskilled (U) or skilled (S)—and can

move from the first to the second by completing training. In both markets firms are homogeneous and jobs are characterised by an idiosyncratic match-specific productivity $p \in (0, 1]$. The key asymmetry is in how match quality evolves: in the unskilled market, new matches begin at the frontier quality $p_0 = 1$ and deteriorate after shocks, whereas in the skilled market quality is drawn from a distribution $\Gamma(\cdot \mid z)$ at formation and redrawn after shocks. The skilled market also supports on-the-job search and a job ladder, giving trained workers richer career dynamics.

Aggregate states

The aggregate state of the economy is $z \in \mathcal{Z}$, where $\mathcal{Z}$ is a finite set. A regime z determines not merely sector productivities but the full constellation of parameters governing each market:

$$z \longmapsto \big\{P_U(z),\, P_S(z),\, \alpha_U(z),\, a_\Gamma(z),\, b_\Gamma(z),\, \lambda_U(z),\, \lambda_S(z),\, \xi_S(z),\, k_U(z),\, k_S(z)\big\}. \tag{1}$$

The sector-specific productivities $P_j(z)$ scale output, the parameters $\alpha_U(z)$, $a_\Gamma(z)$, and $b_\Gamma(z)$ govern the match-quality distributions, $\lambda_j(z)$ are shock arrival rates, $\xi_S(z)$ is the exogenous skilled separation hazard, and $k_j(z)$ are vacancy-posting costs. When the economy switches from one regime to another, all of these objects shift jointly.

At an exogenous date T, the economy switches unexpectedly and permanently from an initial regime z_0 to a new regime $z_1 \neq z_0$.

2.2 Production

A match between a worker of type x and a job with match-specific productivity p in sector $j \in \{U, S\}$ produces flow output

$$\pi_j(x, p, z) = P_j(z) \cdot x \cdot p. \tag{2}$$

Output is multiplicatively separable in ability and match quality. The sector shifter $P_j(z)$ captures aggregate conditions. Training changes market access only: it does not raise innate productivity or generate within-market human capital accumulation.

2.3 Matching

Each market has its own Cobb–Douglas matching function

$$M_j\big(U_j^{\text{eff}}, V_j\big) = \mu_j \left(U_j^{\text{eff}}\right)^{\eta_j} V_j^{1-\eta_j}, \qquad \eta_j \in (0,1). \tag{3}$$

In the unskilled market only unemployed workers search, so $U_U^{\text{eff}} = U_U$. In the skilled market employed workers may also search on the job, so the effective pool of seekers is $U_S^{\text{eff}}(z) = U_S(z) + \int\!\!\int s^*(x,p,z)\, e_S(x,p,z)\, dp\, dx$, where $s^* \in \{0,1\}$ is the on-the-job search decision and e_S is the density of skilled employed. Market tightness is $\theta_j(z) = V_j(z)/U_j^{\text{eff}}(z)$, the vacancy-filling rate is $q_j(\theta_j) = \mu_j \theta_j^{-\eta_j}$, and the job-finding rate for a seeker is $\theta_j(z)\, q_j(\theta_j(z))$. Throughout I write $\kappa_j(z) \equiv \theta_j(z)\, q_j(\theta_j(z))$ for the contact rate in market j: the rate at which any active job seeker—whether unemployed or searching on the job—meets a vacancy. These contact rates are the key prices through which labour-market conditions feed into training decisions.

2.4 Training

Only untrained workers who are currently unemployed in the unskilled market may start training. Doing so requires a one-time utility cost $c(x) = c_0\,(1-x)\,e^{-x}$ that declines in ability. Once a worker begins training, completion arrives at Poisson rate $\phi > 0$, at which point the worker enters skilled unemployment.

While out of work, workers receive flow payoffs that depend on their status: b_U for unskilled unemployed, b_T for those in training, and b_S for skilled unemployed. These payoffs represent the value of non-employment to the worker—a composite of home production, informal income, and the amenity value of leisure. They are structural parameters, constant across aggregate states.[1]

The value of being in training satisfies

$$(r + \phi + \nu)\, T(x,z) = b_T + \phi\, U_S(x,z), \tag{4}$$

so it depends on the value of skilled unemployment $U_S(x,z)$—the channel through which skilled-market conditions feed into the training decision.

[1] In the U.S., UI eligibility requires prior employment and expires after a fixed duration, so b_j should not be read literally as an unemployment-insurance payment. Rather, b_j captures the average flow value that an unemployed worker in segment j enjoys, inclusive of any UI entitlements, home production, and other non-market income. In the policy experiments of Section **??** I vary b_U and b_S jointly to study the effect of a permanent increase in the generosity of the non-employment safety net.

An entering unskilled unemployed worker of type x faces the choice between searching in the unskilled market and paying the training cost to begin the pipeline toward the skilled market. The value of searching satisfies $(r+\nu)\,U_U^{\text{search}}(x,z) = b_U + \kappa_U(z)[E_U(x,1,z) - U_U^{\text{search}}(x,z)]$, reflecting the flow payoff b_U and the option of contacting a vacancy at the frontier quality $p_0 = 1$. The overall value of unskilled unemployment is then

$$U_U(x,z) = \max\left\{U_U^{\text{search}}(x,z),\ -c(x) + T(x,z)\right\}. \tag{5}$$

Because $c'(x) < 0$ and $T(x,z)$ is increasing in x, the net gain from training is increasing in ability. This yields a cutoff.

Proposition 1 (Training cutoff)**.** *Fix a regime $z \in \mathcal{Z}$. There exists a threshold $\bar{x}(z) \in [0,1]$ such that the optimal training policy is $\tau(x,z) = \mathbf{1}\{x > \bar{x}(z)\}$.*

The cutoff $\bar{x}(z)$ summarises the education margin. It responds to aggregate conditions through three channels: the skilled wage premium, the probability of finding a skilled job after training, and the value of the unskilled outside option.

2.5 Unskilled Labour Market

Match quality and shocks

New unskilled matches begin at the frontier quality $p_0 = 1$, so every hire starts at full productivity. Idiosyncratic damage shocks arrive at Poisson rate $\lambda_U(z)$, after which a new quality is drawn from $G_z(p) = p^{\alpha_U(z)}$. If the new draw falls below a reservation threshold $p_U^*(x,z)$—to be determined in equilibrium—the match is destroyed and the worker returns to unemployment.

Surplus

Nash bargaining with worker share β_U splits the total surplus $S_U(x,p,z) = [E_U(x,p,z) - U_U(x,z)] + J_U(x,p,z)$ in fixed proportions, where E_U is the worker's employment value and J_U is the firm's. Summing the Bellman equations of both parties yields a recursion for the surplus that is independent of the wage:

$$(r+\nu+\lambda_U(z))\,S_U(x,p,z) = P_U(z)\,x\,p - (r+\nu)\,U_U(x,z) + \lambda_U(z)\int_{p_U^*}^{1} S_U(x,p',z)\,dG_z(p'). \tag{6}$$

The surplus equals discounted output minus the flow value of the worker's outside option, plus a continuation term reflecting the option value of surviving a damage shock and drawing

a quality above the reservation threshold.

Reservation quality

The threshold $p_U^*(x, z)$ is defined by $S_U(x, p_U^*, z) = 0$. Higher-ability workers tolerate lower match quality because $P_U(z)\, x\, p$ rises faster in p when x is large. An increase in the outside option—whether through better search conditions or a more attractive training option—raises the threshold and makes fragile matches more likely to dissolve.

Wages

The Nash sharing rule gives the wage

$$w_U(x, p, z) = \beta_U\, P_U(z)\, x\, p + (1 - \beta_U)(r + \nu)\, U_U(x, z). \tag{7}$$

The worker receives a share β_U of flow output plus a flow return proportional to the outside option. Because both terms are linear in p, the wage inherits the linearity of the production function.

Vacancies and free entry

Posting a vacancy costs $k_U(z)$ per unit time and generates contacts at rate $q_U(\theta_U(z))$. Since every new match starts at $p_0 = 1$, free entry requires

$$\frac{k_U(z)}{q_U(\theta_U(z))} = \int_0^1 (1 - \beta_U)\, S_U(x, 1, z)\, \hat{u}_U(x, z)\, dx, \tag{8}$$

where $\hat{u}_U(x, z) = u_U(x, z) / \int u_U(x', z)\, dx'$ is the composition of the unskilled unemployed pool. Because the training cutoff determines which types remain in this pool, the education margin feeds directly into vacancy creation.

2.6 Skilled Labour Market

The skilled market features a job ladder: employed workers search on the job and climb toward better matches. Wages are determined by Nash bargaining in the presence of on-the-job search, following Cahuc et al. (2006).

Match quality and shocks

Skilled matches face two sources of destruction. Exogenous separation occurs at rate $\xi_S(z)$, and match quality is hit by shocks at rate $\lambda_S(z)$. After being shocked, match quality is redrawn from a Beta$(a_\Gamma(z), b_\Gamma(z))$ distribution Γ_z on $[0,1]$ with density γ_z. If the new draw falls below the reservation cutoff $p_S^*(x,z)$, the match is endogenously destroyed. Unlike in the unskilled market, skilled matches can improve or worsen after a shock—a feature that interacts with the job ladder to generate rich mobility dynamics.

On-the-job search

An employed skilled worker can search on the job at flow cost $\sigma_S \geq 0$. Outside offers arrive at rate $\kappa_S(z)$—the same contact rate that skilled unemployed workers face—with each offer an independent draw $\tilde{p} \sim \Gamma_z$. The worker moves to the outside job if and only if it yields a higher value. Workers in weak matches gain the most from search because the expected improvement is large; workers in strong matches may find the flow cost not worth paying. This generates a second cutoff $p_S^{oj}(x,z) \geq p_S^*(x,z)$: workers with $p < p_S^{oj}$ search on the job, workers with $p \geq p_S^{oj}$ do not.

Surplus

For each search regime $s \in \{0,1\}$, define the total surplus as $S_S^s(x,p,z) = [E_S^s(x,p,z) - U_S(x,z)] + J_S^s(x,p,z)$, and let $S_S(x,p,z) = \max\{S_S^0(x,p,z),\ S_S^1(x,p,z)\}$ be the surplus under the optimal search decision.

For a worker who does not search on the job ($s=0$), the surplus satisfies

$$(r+\nu+\xi_S(z)+\lambda_S(z))\, S_S^0(x,p,z) = P_S(z)\, x\, p - (r+\nu)\, U_S(x,z) + \big(\lambda_S(z) - \beta_S\, \kappa_S(z)\big) \int_{p_S^*}^1 S_S(x,p',z)\, d\Gamma_z(p'). \tag{9}$$

The coefficient $\lambda_S(z) - \beta_S\, \kappa_S(z)$ on the integral reflects two forces: quality shocks at rate $\lambda_S(z)$ redraw the match and deliver continuation value, while the worker's outside option embeds the contact rate $\kappa_S(z)$ weighted by the bargaining share β_S.

For a searching worker ($s = 1$), the surplus acquires an additional term from the job ladder:

$$\begin{aligned}
&\big(r + \nu + \xi_S(z) + \lambda_S(z) + \kappa_S(z)(1 - \Gamma_z(p))\big)\, S_S^1(x, p, z) \\
&\quad = P_S(z)\, x\, p - \sigma_S - (r + \nu)\, U_S(x, z) \\
&\qquad + \lambda_S(z) \int_{p_S^*}^{1} S_S(x, p', z)\, d\Gamma_z(p') - \beta_S\, \kappa_S(z) \int_{p_S^*}^{p} S_S(x, p', z)\, d\Gamma_z(p'). \qquad (10)
\end{aligned}$$

The left-hand side includes the poaching hazard $\kappa_S(z)(1-\Gamma_z(p))$: when the worker searches, outside offers that beat the current match quality arrive at rate $\kappa_S(z)$ and are accepted with probability $1 - \Gamma_z(p)$. On the right, the integral from p_S^* to p reflects the worker's partial appropriation of the ladder option through bargaining.

The value of skilled unemployment satisfies

$$(r + \nu)\, U_S(x, z) = b_S + \kappa_S(z) \int_{p_S^*}^{1} \beta_S\, S_S(x, \tilde{p}, z)\, d\Gamma_z(\tilde{p}). \qquad (11)$$

Wage schedules

Nash bargaining with worker share β_S applies in the skilled market as well. The resulting wages differ depending on whether the worker searches. For a non-searching worker,

$$\begin{aligned}
w_S^0(x, p, z) &= \beta_S\, P_S(z)\, x\, p + (1 - \beta_S)\, b_S \\
&\quad + \beta_S(1 - \beta_S)\, \kappa_S(z) \int_{p_S^*}^{1} S_S(x, p', z)\, d\Gamma_z(p'). \qquad (12)
\end{aligned}$$

For a searching worker,

$$\begin{aligned}
w_S^1(x, p, z) &= \beta_S\, P_S(z)\, x\, p + (1 - \beta_S)(b_S + \sigma_S) \\
&\quad + \beta_S(1 - \beta_S)\, \kappa_S(z) \int_{p_S^*}^{p} S_S(x, p', z)\, d\Gamma_z(p'). \qquad (13)
\end{aligned}$$

Workers who search on the job earn strictly less than non-searchers at the same (x, p). Two forces drive the discount: Nash bargaining shifts the flow search cost partly onto the worker, and firms anticipate that a searching worker is more likely to be poached, reducing the expected duration of the match and thereby the surplus the firm is willing to share. Both effects are strongest near the reservation threshold, where the gain from climbing the ladder is largest.

Vacancies and free entry

Posting a skilled vacancy costs $k_S(z)$ per unit time and generates contacts at rate $q_S(\theta_S(z))$. The pool of seekers includes both unemployed and on-the-job searchers, so the expected value of a contact depends on whom the firm meets. Let $\bar{J}_S(z)$ denote this expected value, averaged over the composition of seekers. Free entry requires

$$k_S(z) = q_S(\theta_S(z))\, \bar{J}_S(z), \tag{14}$$

pinning down $\theta_S(z)$. Because the training cutoff $\bar{x}(z)$ determines the inflow into skilled unemployment and the on-the-job search cutoff determines how many employed workers are active on the market, both policy margins feed into the free-entry condition.

2.7 Stationary Equilibrium

Fix a regime $z \in \mathcal{Z}$. A stationary equilibrium consists of value functions (U_U, U_U^{search}, T, E_U, U_S, E_S^0, E_S^1), firm values (J_U, J_S^0, J_S^1), policy rules (τ, p_U^*, p_S^*, p_S^{oj}, s^*), market tightness (θ_U, θ_S), and cross-sectional distributions (u_U, t, u_S, e_S) such that:

(i) **Optimality.** Worker and firm value functions satisfy their Bellman equations; training, separation, and on-the-job search rules are optimal given those values.

(ii) **Wage determination.** Wages satisfy the Nash-bargaining rules in equations (7), (12), and (13).

(iii) **Free entry.** Tightness satisfies the zero-profit conditions (8) and (14).

(iv) **Stationarity.** Cross-sectional distributions satisfy flow-balance conditions, with inflows equal to outflows in every state.

(v) **Consistency across markets.** The completion flow from training equals the inflow into skilled unemployment.

The two markets are linked through the training cutoff: it depends on the value of skilled unemployment, which depends on skilled-market tightness, which depends on the distribution of skilled workers, which depends on the training inflow. I solve for the equilibrium with a nested fixed-point algorithm described in Appendix A.

2.8 Model Properties

The structure delivers several properties that guide the empirical and policy analysis below.

Training cutoff. The cutoff $\bar{x}(z)$ responds to aggregate conditions through three channels: changes in the skilled wage premium, changes in the probability of finding a skilled job after training, and changes in the value of the unskilled outside option. A regime that raises $P_S(z)$ relative to $P_U(z)$ lowers the cutoff and pushes more workers into training; a regime that raises the skilled contact rate $\kappa_S(z)$ has the same effect by increasing the arrival rate of skilled jobs.

Endogenous separation. Higher-ability workers tolerate lower match quality in both markets, because the flow surplus $P_j(z)\, x\, p$ rises faster in p when x is large. An increase in the outside option—whether through improved search conditions or a more attractive training option—raises reservation thresholds and makes fragile matches more likely to dissolve.

On-the-job search and wage dispersion. The search cutoff $p_S^{oj}(x, z)$ generates wage dispersion in the skilled market even among workers of identical ability. Workers who have been lucky enough to climb to high-quality matches earn more than those still searching, purely because of the frictional process of job-ladder climbing.

The aggregate state. Because z shifts $\lambda_S(z)$, $\xi_S(z)$, and the match-quality distribution Γ_z jointly with $P_S(z)$, a recession is not merely a productivity shortfall: it also changes the pace at which workers are displaced and the rate at which match quality deteriorates. Separating these channels is the main motivation for the structural approach in Section 3.

3 Data and Estimation

3.1 Data

The model is estimated on U.S. data from four sources. Labour-market stocks and worker transition rates are constructed from the Current Population Survey—Basic Monthly Files (CPS). Wage moments are drawn from the CPS Annual Social and Economic Supplement (ASEC). The skilled employer-to-employer transition rate comes from the Census Bureau's Job-to-Job Flows (J2J) data, which provides quarterly counts of hires from other employers by education group. Vacancy data for market tightness come from the Job Openings and Labor Turnover Survey (JOLTS).

The sample consists of people (civilians, military was excluded) aged 25–54 who report positive labour-force attachment. I partition workers into two skill groups: those without a bachelor's degree (unskilled) and those with a bachelor's degree or above (skilled). This

partition is the empirical counterpart of the model's two-market structure. Monthly transition rates—unemployment to employment (UE), employment to unemployment (EU), and employer to employer (EE)—are constructed from matched CPS records using standard methods.

Moments are computed over four windows, each chosen to approximate a stationary regime:

Table 1: Estimation windows

Period	Dates	Length
Baseline fc	Jan 2003 – Dec 2007	60 months
Crisis fc	Jan 2008 – Jun 2009	18 months
Baseline covid	Jan 2015 – Dec 2019	60 months
Crisis covid	Mar 2020 – Dec 2021	22 months

The two baseline windows cover the expansion phases preceding each crisis. The crisis windows cover the contraction through the trough. Within each window, I average the monthly observations to obtain a single vector of target moments and treat this average as the stationary equilibrium that the model should match.

3.2 Moments and Identification

I estimate the model by the Simulated Method of Moments in a standard fashion. The estimation targets 23 moments grouped into four blocks. Table 2 lists each moment with its model counterpart.

The choice of moments reflects the structure of the model. The skilled share and training share directly discipline the training cutoff $\bar{x}(z)$. The cross-sectional wage dispersion moments—variance and skewness of the wage distribution within each skill group—pin down the shape parameters of the match-quality distributions and help separate the ability distribution from the match-quality distribution. Without these moments, the two sources of wage variation would be poorly identified separately.

Transition rates play the central role in identifying market-specific frictions. The unskilled UE rate identifies the contact rate $\kappa_U(z)$, and therefore—through the matching function and free-entry condition—the matching efficiency and vacancy cost. The unskilled EU rate identifies the shock arrival rate $\lambda_U(z)$ and the reservation quality $p_U^*(x, z)$: given the production function, these two objects are tightly linked. On the skilled side, the same logic applies, with the additional information from the employer-to-employer (EE) transition rate. This moment is particularly valuable because it identifies the intensity of on-the-job search: it

Table 2: Targeted moments

Moment	Model counterpart	Source
Labour-market stocks		
Skilled share	$\int_{\bar{x}}^{1} [u_S(x) + \int e_S(x,p)\, dp]\, dx$	CPS
Training share	$\int_{\bar{x}}^{1} t(x)\, dx$	CPS
Variance of unskilled wages	$\mathrm{Var}[w_U]$	CPS ASEC
Skewness of unskilled wages	$\mathrm{CM}_3[w_U]$	CPS ASEC
Variance of skilled wages	$\mathrm{Var}[w_S]$	CPS ASEC
Skewness of skilled wages	$\mathrm{CM}_3[w_S]$	CPS ASEC
Transition rates		
Job-finding rate, unskilled	$\kappa_U(z)\,[1 - G_z(p_U^*)]$	CPS
Separation rate, unskilled	$\lambda_U(z)\, G_z(p_U^*)$	CPS
Job-finding rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^*)]$	CPS
Separation rate, skilled	$\xi_S(z) + \lambda_S(z)\,\Gamma_z(p_S^*)$	CPS
EE rate, skilled	$\kappa_S(z)\,[1 - \Gamma_z(p_S^{oj})]$	J2J
Wages		
Mean wage, unskilled	$\mathbb{E}[w_U(x,p,z)]$	CPS ASEC
Mean wage, skilled	$\mathbb{E}[w_S(x,p,z)]$	CPS ASEC
25th percentile wage, unskilled	$Q_{0.25}[w_U]$	CPS ASEC
25th percentile wage, skilled	$Q_{0.25}[w_S]$	CPS ASEC
Median wage, unskilled	$Q_{0.50}[w_U]$	CPS ASEC
Median wage, skilled	$Q_{0.50}[w_S]$	CPS ASEC
Log skill premium	$\mathbb{E}[\ln w_S] - \mathbb{E}[\ln w_U]$	CPS ASEC
Tightness		
θ_U	V_U/U_U	JOLTS / CPS
θ_S	V_S/U_S^{eff}	JOLTS / CPS

pins down the contact rate $\kappa_S(z)$ jointly with the on-the-job search cutoff $p_S^{oj}(x, z)$ and helps separate the offer distribution Γ_z from the separation threshold $p_S^*(x, z)$.

Wage levels identify the productivity shifters $P_U(z)$ and $P_S(z)$. The log skill premium identifies the return to training. Wage quantiles—particularly the gap between the 25th percentile and the median—are informative about the shape of the match-quality distributions and the bargaining shares β_j: a compressed lower tail relative to the median suggests a tight match-quality distribution or high bargaining power, while a long lower tail suggests the opposite.

Market tightness $\theta_j(z)$ directly constrains vacancy costs $k_j(z)$ through the free-entry conditions. Without these moments, vacancy costs would be identified only indirectly through their effect on contact rates, which also depend on matching efficiencies.

3.3 Estimation

The SMM estimator minimises the weighted distance between the data moments $\hat{m}$ and their model counterparts $m(\theta)$:

$$\hat{\theta} = \arg\min_{\theta} \big[\hat{m} - m(\theta)\big]' \, W \, \big[\hat{m} - m(\theta)\big], \tag{15}$$

where W is the optimal weight matrix, constructed as the inverse of the estimated variance–covariance matrix of $\hat{m}$. I estimate $\hat{\Sigma}$ using influence functions: for each time period t in a given window, I compute the vector of moment deviations $\psi_t = m_t - \hat{m}$, stack these into a $T \times K$ matrix Ψ, and form $\hat{\Sigma} = \Psi'\Psi/T$. Each column of Ψ is scaled by the absolute value of the corresponding moment to ensure scale invariance. When $\hat{\Sigma}$ is ill-conditioned, off-diagonal elements are shrunk toward zero until the condition number falls below a threshold. In practice, multiplying all of diagonal elements by a shrinkage rate between 0.95 and 0.99 was enough to stabilise the matrix.

The model moments $m(\theta)$ are computed by solving for the stationary equilibrium at parameter vector θ and evaluating the equilibrium objects described in Section 2.

Parameter classification

Three parameters—the discount rate r, the demographic exit rate ν, and the training completion rate ϕ—are calibrated externally and held fixed across all estimation windows. Table 3 reports their values.

A second group of parameters is fixed externally to simplify the model. Firms and workers share surplus equally and the matching function depends symmetrically on unemployment

Table 3: Externally calibrated parameters

Parameter	Source	Value
r	Monthly discount rate ($\approx 5\%$ annual)	0.0042
ν	CPS matched-panel labour-force exit hazard	0.0384
ϕ	National Student Clearinghouse completion rate	0.0222

and vacancies, so the bargaining shares (β_U, β_S) and matching elasticities (η_U, η_S) are all set to 0.5. The OJS flow cost is likewise fixed, at $\sigma_S = 1.1$.

The remaining parameters are estimated, and fall into two groups. *Deep structural parameters* reflect preferences, institutions, and technology that do not change with the aggregate state: the ability distribution (a_ℓ, b_ℓ), the training cost c, the flow payoffs (b_U, b_T, b_S), and the matching efficiencies (μ_U, μ_S). *Regime-specific parameters* capture features of the aggregate state that shift between regimes: the sector productivities (P_U, P_S), the match-quality distribution parameters $(\alpha_U, a_\Gamma, b_\Gamma)$, the vacancy costs (k_U, k_S), the shock arrival rates (λ_U, λ_S), and the exogenous skilled separation hazard ξ_S.

Multi-regime estimation strategy

The estimation proceeds in two stages. In the first stage, I estimate the model on the baseline window, treating all deep structural and regime-specific parameters as free. This stage identifies the full parameter vector from a single cross-section of moments.

In the second stage, I fix the deep structural parameters at their baseline estimates and re-estimate only the regime-specific parameters on the crisis window. This procedure identifies what changed during the crisis: which parameters shifted, by how much, and in which direction. The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the baseline and the crisis—only the aggregate state did.

The same two-stage procedure is applied separately to the financial-crisis moments pair (baseline fc $\rightarrow$ crisis fc) and the COVID pair (baseline covid $\rightarrow$ crisis covid). Comparing the deep structural estimates across the two baselines provides a structural stability check: if the deep parameters are similar across the two expansions, the identifying assumption that they are regime-invariant is internally consistent.

4 The Financial Crisis

This section presents the estimation results for the financial-crisis episode. I begin with the baseline equilibrium estimated on the pre-crisis expansion (2003–2007), examine the model's internal structure through the lens of the estimated parameters, assess the quality of the moment fit, and then turn to the crisis re-estimation (2008–2009) to identify which features of the aggregate state shifted during the downturn.

4.1 Baseline Equilibrium

The model is estimated on the baseline fc window and solved at the estimated parameters. I now describe the resulting equilibrium objects.

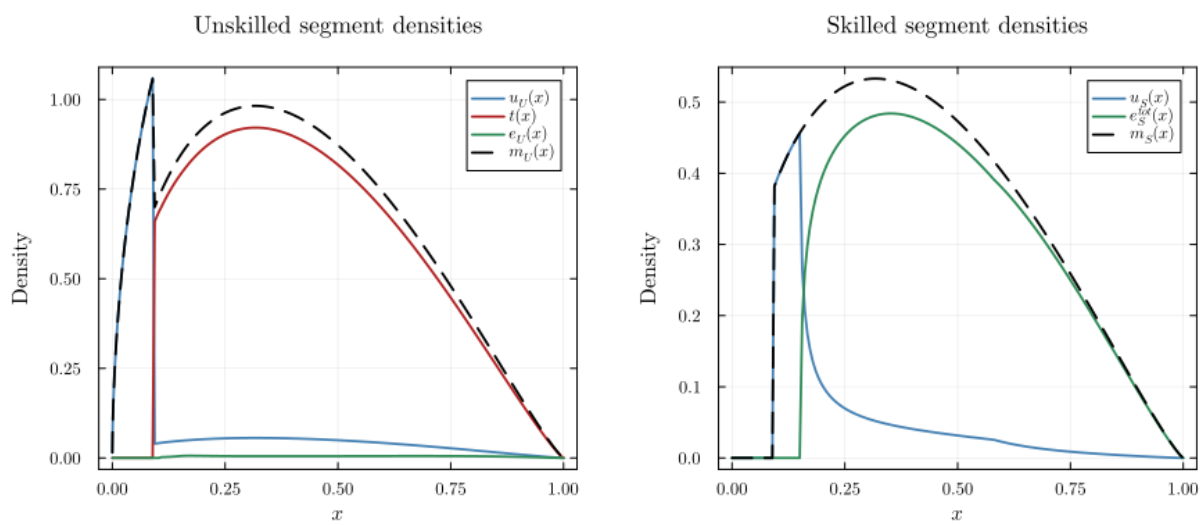

Figure 1: Cross-sectional densities over worker type x: unskilled segment (left) and skilled segment (right). The training cutoff $\bar{x}$ partitions the two markets. See text for details.

Cross-sectional distributions. *Discussion:* The figure shows the endogenous segmentation of the economy. The training cutoff partitions the ability distribution into two pools with different mean characteristics. Above $\bar{x}$, the mass flows through the training pipeline into skilled unemployment and then into skilled employment. Below $\bar{x}$, workers remain in the unskilled market permanently.

Training decision. *Discussion:* For low-ability workers, the value of unskilled search exceeds the net value of training: the training cost is too high and the expected gains from the skilled market too low. For high-ability workers, the net value of training dominates:

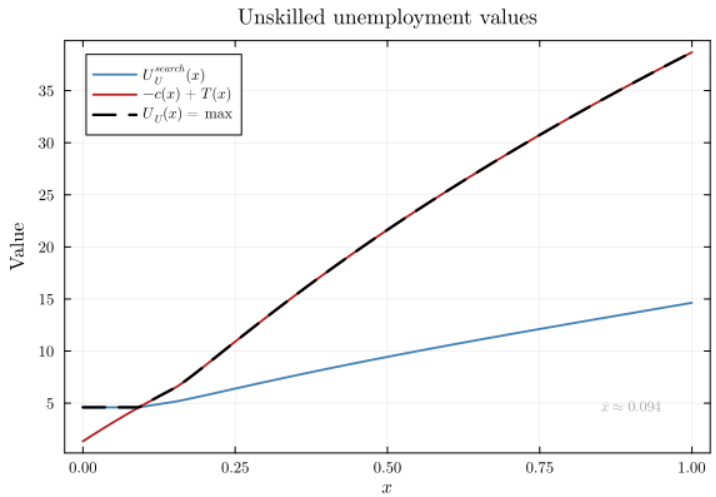

Figure 2: Unskilled unemployment values. The value of unskilled search $U_U^{\text{search}}(x)$, the net value of training $-c(x)+T(x)$, and the upper envelope $U_U(x)$ are plotted over worker type x. Their intersection defines the training cutoff $\bar{x}$.

training costs are low, and the continuation value of the skilled market—including job-ladder gains from on-the-job search—is high. The gap between the two curves at $\bar{x}$ is zero by construction; above $\bar{x}$ it widens, reflecting the increasing net gain from training for higher-ability types.

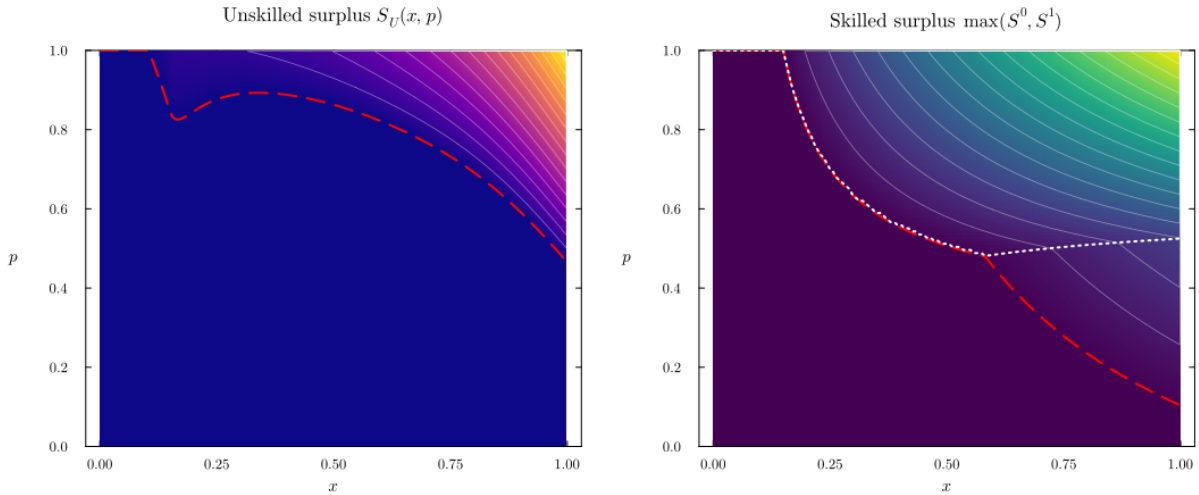

Figure 3: Match surplus over the (x, p) plane: unskilled surplus $S_U(x, p)$ (left) and skilled surplus $\max(S_S^0, S_S^1)$ (right). Overlaid curves mark the reservation quality $p_j^*(x)$ and, in the skilled panel, the on-the-job search cutoff $p_S^{oj}(x)$.

Surplus and separation rules. *Discussion:* The surplus maps summarise the equilibrium in each market. The key comparative static is the slope of the reservation cutoff in x: flatter

curves mean that the training threshold has a weaker effect on match stability.

4.2 Moment Fit

Note: Results are not ready yet for the following sections!

The quality of the baseline estimation is assessed by comparing data moments $\hat{m}$ with their model counterparts $m(\hat{\theta})$. A scatter plot places data values on the horizontal axis and model values on the vertical axis, with a 45-degree line marking perfect fit. Points are grouped by moment category: labour-market stocks, transition rates, wages, and tightness. Systematic deviations—if any—are informative about model limitations, such as the absence of within-market human capital accumulation or the assumption of homogeneous firms. Stock moments are typically matched more tightly than flow moments, because the stationary distributions integrate over idiosyncratic shocks and therefore average out high-frequency noise.

4.3 Parameter Estimates

The full set of parameter estimates is reported in a single table spanning all four estimation windows. The table has three blocks. The first block lists the externally calibrated parameters (r, ν, ϕ), which are identical across all columns. The second block lists the deep structural parameters—the ability distribution (a_ℓ, b_ℓ), the training cost c, the flow payoffs (b_U, b_T, b_S), and the matching efficiencies (μ_U, μ_S)—estimated in the baseline columns and held fixed at those values in the crisis columns. The two baseline columns (pre-FC and pre-COVID) allow a comparison across expansions as a structural stability check. The third block lists the ten regime-specific parameters, estimated freely in all four columns. The Δ columns report the change from baseline to crisis and constitute the main object of interest.

4.4 Crisis Re-estimation

The second-stage estimation fixes the deep structural parameters at their baseline fc values and re-estimates the ten regime-specific parameters on the crisis fc window (January 2008 – June 2009). The identifying assumption is that preferences, matching technology, and bargaining institutions did not change between the expansion and the recession—only the aggregate state did.

The key questions the re-estimation answers are: Did sector productivities P_U and P_S fall, reflecting the direct output loss from the recession? Did the shock arrival rates λ_U and λ_S

rise, indicating that matches were hit by adverse productivity draws more frequently? Did vacancy costs k_U and k_S rise, reflecting tighter credit conditions or increased uncertainty that discouraged hiring? Did the exogenous skilled separation hazard ξ_S change, indicating layoffs unrelated to match-specific productivity? And how did the match-quality distribution parameters $(\alpha_U, a_\Gamma, b_\Gamma)$ shift—did the composition of available jobs change during the crisis?

The financial crisis is primarily a demand shock. The expected pattern is a broad decline in productivities, a rise in vacancy costs, and a moderate increase in shock arrival rates. The non-employment flow values b_U and b_T should remain roughly stable, since standard unemployment insurance was not dramatically expanded during the Great Recession.

4.5 Training Cutoff

The training cutoff under the crisis parameters, $\bar{x}(z_1)$, is compared with its baseline counterpart $\bar{x}(z_0)$. A lower cutoff during the crisis means that more workers find it optimal to train—the model generates countercyclical enrolment endogenously. The mechanism operates through the unskilled contact rate $\kappa_U(z)$: a decline in this rate compresses the continuation value of remaining unskilled and makes training relatively more attractive. This connects to the empirical literature documenting countercyclical enrolment (Dellas and Sakellaris, 2003; Barr and Turner, 2015; Blom et al., 2021).

Whether the cutoff actually falls during the financial crisis depends on the relative severity of the demand shock across the two markets. If both skilled and unskilled prospects worsen roughly symmetrically, the net effect on the training margin is ambiguous: the unskilled outside option deteriorates, but so does the value of the skilled market that workers are training toward. The estimation will reveal which force dominates.

4.6 Transition Dynamics

The transition-dynamics methodology developed in Section **??** is applied to the financial-crisis regime switch. At the moment of the switch, regime-specific primitives jump to their crisis values but the cross-sectional distributions are predetermined, inherited from the pre-crisis steady state. The economy then follows a joint forward equilibrium path as tightness, policy rules, and distributions co-evolve toward the new stationary equilibrium.

The transition exercise reports time paths for unemployment rates $\mathcal{U}_U(t)$ and $\mathcal{U}_S(t)$, the training stock $\mathcal{T}(t)$, market tightness $\theta_U(t)$ and $\theta_S(t)$, the training cutoff $\bar{x}(t)$, the skilled share, and mean wages by sector. The path of the training cutoff is of particular interest: it may overshoot its new steady-state value on impact—when the distributions have

not yet adjusted—and then partially revert as the composition of the unemployment pool changes. Such non-monotonicity would indicate that the general-equilibrium feedback from the training inflow on skilled-market tightness operates with a lag.

5 COVID-19

5.1 Pre-COVID Baseline (2015–2019)

The structural stability check fixes the deep parameters at their financial-crisis baseline estimates and re-estimates only the regime-specific parameters on the pre-COVID expansion (January 2015 – December 2019). If the deep parameters identified from the 2003–2007 expansion can rationalise a second, independent expansion twelve years later—with only the aggregate state allowed to differ—the identifying assumption that these parameters are regime-invariant gains substantial credibility. The moment fit for the pre-COVID baseline is reported alongside the financial-crisis baseline fit to facilitate comparison.

5.2 Crisis Estimation (2020–2021)

The second-stage estimation is repeated for the COVID window (March 2020 – December 2021), again fixing the deep parameters and re-estimating the regime-specific parameters. COVID differs qualitatively from the financial crisis: it is a supply and reallocation shock rather than a demand shock. The expected pattern of parameter shifts reflects this distinction.

Unskilled productivity P_U is expected to fall sharply, since contact-intensive service occupations—disproportionately staffed by workers without a bachelor's degree—were hit hardest by lockdowns and social distancing. Skilled productivity P_S should fall by less, reflecting remote-work compatibility in many professional occupations. The exogenous skilled separation hazard ξ_S is expected to spike, capturing the mass temporary layoffs that affected even high-skill workers during the initial shutdown. The shock arrival rate λ_U should rise sharply, reflecting massive job destruction in the service sector.

A distinctive feature of the COVID episode is the expansion of non-employment benefits. The FPUC supplement and Pandemic Unemployment Assistance substantially raised the flow value of non-employment. In the model, this is captured by increases in b_U and b_T. Because b_T enters the training value directly through equation (4), a higher non-employment payoff during training lowers the effective cost of the training spell and reinforces any incentive to retrain.

5.3 Training Cutoff

The COVID training cutoff $\bar{x}(z_{\text{COVID}})$ is compared with its pre-COVID baseline. The hypothesis is that the cutoff rises—that is, more workers find it optimal to train—because the unskilled sector is disproportionately damaged while expanded UI lowers the effective cost of the training spell, making the skilled sector relatively more attractive. This provides a structural explanation for the observed enrolment spike documented by Barr and Turner (2015) and subsequent studies of pandemic-era education decisions.

The contrast with the financial crisis is central. If the financial crisis is a roughly symmetric demand shock that leaves the training margin ambiguous, COVID is an asymmetric supply shock that tilts it decisively toward retraining. The direction-of-change comparison is the paper's main empirical prediction.

5.4 Transition Dynamics

The same transition exercise is applied starting from the pre-COVID steady state. The COVID transition is expected to differ from the financial-crisis transition in its initial dynamics: a sharp spike in unskilled unemployment driven by the separation-rate shock, with partial reversal as temporary layoffs are recalled. The training cutoff may jump upward on impact—as the unskilled outside option collapses—and then gradually settle toward its new steady-state value as the expanded UI benefits and the recomposition of the unemployment pool work through the equilibrium.

6 Comparing the Two Crises

With both crises estimated under the same deep structure, a systematic comparison is possible.

6.1 Parameter Shifts

A side-by-side table reports the regime-specific parameter changes $\Delta\theta^{\text{FC}}$ and $\Delta\theta^{\text{COVID}}$ for each of the ten parameters. The questions to answer are: which parameters move in the same direction across both crises, and which move differently? Is the financial crisis primarily a "vacancy-cost shock"—elevated k_U, k_S reflecting credit-market tightening—while COVID is a "separation-rate + UI shock"—elevated ξ_S, λ_U, b_U, b_T reflecting mass layoffs and expanded benefits? Differences in the pattern of parameter shifts provide the structural basis for the divergent training-cutoff predictions.

6.2 Training-Cutoff Comparison

The stationary-regime cutoffs $\bar{x}^{ss}(z_{\text{FC}})$ and $\bar{x}^{ss}(z_{\text{COVID}})$ are compared directly. If the two crises generate opposite-sign movements in the training cutoff, the model produces a sharp testable implication: demand shocks and supply shocks have qualitatively different effects on human-capital investment. A decomposition attributes the total cutoff change to contributions from each parameter shift, identifying which channels—productivity, contact rates, separation rates, non-employment benefits—drive the difference.

6.3 Transition-Dynamics Comparison

Overlaying the two transition paths reveals differences in the speed of adjustment, overshooting and undershooting patterns, and which crisis generates longer-lasting hysteresis in the skilled share. A fast convergence of the training cutoff to its new steady state would indicate that the general-equilibrium feedbacks are quantitatively weak; a slow, non-monotone path would suggest that compositional effects—the gradual reallocation of workers across markets—play a first-order role in the adjustment.

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A Appendix A

Appendices go here.